

Chemical Equilibrium

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Chemical equilibrium is a dynamic state where the rate of the forward reaction equals the rate of the reverse reaction; concentrations of reactants and products remain constant (but not necessarily equal).
- Law of Mass Action: for $a A + b B \rightleftharpoons c C + d D$, the equilibrium constant $K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$ at a given temperature.
- K_p is the equilibrium constant in terms of partial pressures for gaseous reactions; $K_p = K_c (RT)^{\Delta n}$, where $\Delta n = \text{moles of gaseous products} - \text{moles of gaseous reactants}$.
- Le Chatelier's Principle: if a system at equilibrium is disturbed (by change in concentration, pressure, volume, or temperature), the equilibrium shifts to counteract the disturbance.
- Increasing pressure (decreasing volume) shifts equilibrium toward the side with fewer moles of gas. Adding an inert gas at constant volume does not shift equilibrium.
- A large $K_c (>>1)$ indicates the reaction favors products at equilibrium; a small $K_c (<<1)$ indicates it favors reactants.
- Catalysts speed up both forward and reverse reactions equally, so they help reach equilibrium faster but do not change the equilibrium position (K_c unchanged).



Equilibrium: $\text{rate}(\text{forward}) = \text{rate}(\text{reverse})$

A reversible reaction reaching dynamic equilibrium: forward rate equals reverse rate.

2. Formula Sheet

Equilibrium constant (K_c)	$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$
Relation between K_p and K_c	$K_p = K_c (RT)^{\Delta n}$
Reaction quotient	$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$ (at any time)
Condition for equilibrium	$Q = K_c$
Degree of dissociation	$\alpha = \frac{\text{moles dissociated}}{\text{initial moles}}$
K_c for reverse reaction	$K_c(\text{reverse}) = \frac{1}{K_c(\text{forward})}$
K_c for a reaction multiplied by n	$K_c(\text{new}) = (K_c)^n$
Van't Hoff equation (qualitative)	$\ln(K_2/K_1) = -\Delta H/R (1/T_2 - 1/T_1)$

3. Practice Problems (22 questions)

- For the reaction $A + B \rightleftharpoons C + D$ at equilibrium, $[A]=1M$, $[B]=1M$, $[C]=2M$, $[D]=2M$. Find K_c .
- State Le Chatelier's principle and predict the effect of increasing pressure on $N_2 + 3H_2 \rightleftharpoons 2NH_3$.
- For the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$, find $\Delta n(\text{gas})$ and hence the relation between K_p and K_c .
- Explain why adding a catalyst does not change the value of K_c for a reaction.
- For a reaction, $K_c = 100$ at a certain temperature. State whether the reaction favors products or reactants at equilibrium.
- Predict the effect of increasing temperature on an exothermic reaction at equilibrium, using Le Chatelier's principle.
- For the reaction $2SO_2 + O_2 \rightleftharpoons 2SO_3$, if K_c for the forward reaction is 4, find K_c for the reverse reaction.
- Explain the difference between the reaction quotient Q and the equilibrium constant K .
- For the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, if the degree of dissociation is 0.2 and initial moles of PCl_5 is 1, find the moles of each species at equilibrium.
- State why the equilibrium constant depends only on temperature and not on the initial concentrations of reactants.
- For a reaction with $\Delta n(\text{gas}) = 0$, state the relationship between K_p and K_c .
- Explain the effect of adding an inert gas at constant volume on a gaseous equilibrium.
- For the reaction $2A \rightleftharpoons B$, if $K_c = 0.25$, and $[A] = 2 M$, find $[B]$ at equilibrium.
- State two characteristics of a system at chemical equilibrium.
- Explain, using Le Chatelier's principle, why increasing the concentration of a reactant shifts equilibrium toward the products.
- For a reaction, $K_c(\text{forward}) = 25$. Find K_c for the same reaction written with all coefficients doubled.
- Explain why chemical equilibrium is described as 'dynamic' rather than static.

18. For the reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$, $\Delta n(\text{gas}) = 0$. Explain how a change in total pressure (by changing volume) affects this equilibrium.
19. State the effect of decreasing temperature on an endothermic reaction at equilibrium.
20. For a weak acid dissociation $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$, if $K_a = 1.8 \times 10^{-5}$ and initial concentration is 0.1 M, find the approximate degree of dissociation (using $\alpha = \sqrt{K_a/C}$ for small α).
21. Explain why K_c for a reaction and K_c for its reverse reaction are reciprocals of each other.
22. State how the equilibrium constant of a reaction at a given temperature relates to the standard Gibbs free energy change ($\Delta G^\circ = -RT \ln K$), conceptually.
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Answer Key

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| 1. $K_c = 4$ | 2. Conceptual - equilibrium shifts toward fewer moles of gas (right, toward NH_3) when pressure increases |
| 3. $\Delta n = 2 - 4 = -2$; $K_p = K_c(RT)^{-2}$ | 4. Conceptual - catalyst speeds up forward and reverse reactions equally, so equilibrium position (K_c) is unaffected |
| 5. Favors products ($K_c \gg 1$) | 6. Conceptual - equilibrium shifts toward reactants (since forward is exothermic, increasing T favors the endothermic reverse direction) |
| 7. $K_c(\text{reverse}) = 1/4 = 0.25$ | 8. Conceptual - Q is calculated at any point in the reaction; K is Q's specific value at equilibrium |
| 9. $\text{PCl}_5 = 0.8 \text{ mol}$, $\text{PCl}_3 = 0.2 \text{ mol}$, $\text{Cl}_2 = 0.2 \text{ mol}$ | 10. Conceptual - K_c is derived from the law of mass action which inherently depends only on temperature (via rate constants) |
| 11. $K_p = K_c$ | 12. Conceptual - no effect on equilibrium position at constant volume, since partial pressures of reactants/products don't change |
| 13. $[\text{B}] = 1 \text{ M}$ | 14. Conceptual - forward rate = reverse rate; concentrations remain constant over time |
| 15. Conceptual - adding reactant increases Q below K, so system shifts forward to restore $Q=K$, forming more product | 16. $K_c(\text{new}) = 25^2 = 625$ |
| 17. Conceptual - forward and reverse reactions continue to occur at equal rates, even though net concentrations don't change | 18. Conceptual - little to no shift, since $\Delta n=0$ means partial pressures ratio is unaffected by volume/pressure change |
| 19. Conceptual - equilibrium shifts toward products (forward, endothermic direction reversed) - actually shifts toward reactants since decreasing T favors exothermic reverse direction | 20. $\alpha \approx 0.0134$ (1.34%) |
| 21. Conceptual - forward and reverse K_c expressions are inverses of each other by definition (numerator/denominator swap) | 22. Conceptual - large positive ΔG° standard gives small K (favors reactants); large negative ΔG° standard gives large K (favors products) |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.